

Project Idea

This project is an AI-powered LinkedIn Post Generator that creates professional posts automatically using LLM (Large Language Model).

Using: Groq API key

Model: "llama-3.3-70b-versatile"

User selects:

- **Topic (Tag)** → Example: Job Search, Motivation, Mental Health
- **Length** → Short / Medium / Long
- **Language** → English or Hinglish

Then the system generates a LinkedIn-style post similar to real viral posts.

Working Flow (Step-by-step)

Step 1: Raw Data Collection (raw_posts.json)

You have a file raw_posts.json which contains real LinkedIn posts. These posts are used as training examples.

Which contains text and Engagement in json formats.

Step 2: Preprocessing using LLM (preprocess.py)

Each post is sent to AI model (LLM) which extracts:

- Number of lines
- Language (English / Hinglish)
- Tags (Topic)

Input post: Looking for jobs on LinkedIn is like online dating...

```
Output metadata:{  
  "line_count": 1,  
  "language": "English",  
  "tags": ["Job Search"]  
}
```

get_unified_tags(): Combines similar tags.

Example:Motivation
Inspiration
Drive

becomes: Motivation

Final processed data saved in: processed_posts.json. Which contains text,engagement, line_count, language, tags.

Flow:

```
raw_posts.json
↓
extract metadata
↓
unify tags
↓
processed_posts.json
```

llm_helper.py

Connects project to AI model.

Uses:

ChatGroq
model = llama-3.3-70b-versatile

Loads API key from:

.env file

Example:

```
llm.invoke("What are ingredients in samosa?")
```

Step 3: Organizing posts using Pandas (few_shot.py)

This file:

- Reads processed posts
- Stores them in dataframe
- Categorizes post length:
 - Short → 1-5 lines

- Medium → 6-10 lines
 - Long → 11+ lines
- Extracts unique tags

Example tags: Motivation

Job Search

Mental Health

Scams

Career Growth

`get_filtered_posts()`

Filters posts based on:

- length
- language
- tag

Example:

`get_filtered_posts("Medium","English","Motivation")`

Returns similar posts.

`get_tags()`

Returns all available topics.

Example:

`["Motivation","Job Search","Scams"]`

Step 4: AI generates new post (post_generator.py)

Responsible for generating final post.

`get_length_str()`

Converts:

Short → 1-5 lines

Medium → 6-10 lines

Long → 11-15 lines

`get_prompt()`

Creates final prompt sent to AI.

Example prompt:

Generate LinkedIn post

Topic: Motivation

Length: 6-10 lines

Language: English

Use writing style like examples:

Example 1:

Believe in yourself...

`generate_post()`

Calls LLM:

`llm.invoke(prompt)`

Returns generated post.